

Rule-Based Personalized Meal Recommendation System Using InBody Data

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Keywords: InBody, body composition, personalized meal recommendation, digital healthcare, nutrient calculation algorithm

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I. INTRODUCTION

The increasing prevalence of obesity, together with the associated rise in the risk of metabolic and cardiovascular diseases, has further highlighted the importance of systematic body-composition and nutritional management. Because body weight alone cannot adequately describe an individual's health status, more precise management based on detailed body-composition indicators, such as skeletal muscle mass, body-fat percentage, and basal metabolic rate (BMR), is increasingly necessary.

Alongside the widespread adoption of smartphones, various digital healthcare applications have emerged to provide dietary logging, calorie calculation, and exercise tracking. However, many services depend on users' self-reported data and offer uniform guidance that is limited to simple calorie totals. These approaches are susceptible to data-entry errors and missing records and have difficulty providing personalized meal plans that can be finely adjusted in response to changes in body composition. Body-composition data measured using InBody devices reflect health status more accurately, but in practice they are often used only to review a result sheet or view a simple visualization.

This study proposes a personalized meal recommendation system that directly links InBody-based body-composition data to meal planning. Users enter either their InBody measurements or basic physical information and health goals. The system then calculates recommended daily energy and macronutrient intake and automatically recommends breakfast, lunch, and dinner from a food database. Its objective is to provide practical personalized meal plans that jointly consider body composition, activity level, goals, allergies, and budget.

II. SYSTEM DESIGN AND METHODOLOGY

The proposed system consists of four principal components: InBody integration and body-composition processing, nutrient calculation, food-data collection and management, and meal recommendation with repetition checking. Through the application interface, users enter either their InBody results or basic information such as sex, age, height, and weight. They also specify an objective, such as weight loss or muscle gain, together with their activity level and target period.

During InBody integration and body-composition processing, the system directly uses skeletal muscle mass, body-fat mass, body-fat percentage, BMR, and other measurements when InBody data are available. When these measurements are unavailable, predictive equations based on basic physical information are used to approximate BMR and body-fat percentage. This approach is intended to provide basic personalized meal recommendations both to users who can directly access InBody measurements and to those who cannot.

The nutrient-calculation module uses the Mifflin-St Jeor equation, one of the commonly used predictive equations, to estimate BMR. The resulting BMR is multiplied by an activity factor to obtain total daily energy expenditure, after which the recommended intake is adjusted according to the user's objective. For weight loss, the energy target is set moderately below total expenditure; for weight maintenance or muscle gain, it is set at maintenance level or at a modest surplus. The macronutrient ratio is then configured with a somewhat higher proportion of protein for weight loss and with adequate carbohydrate and protein intake for muscle gain. Daily target amounts are calculated using energy densities of 4 kcal/g for carbohydrates and protein and 9 kcal/g for fat. These values provide the basis for meal recommendation.

The food-data collection and management module is a core determinant of recommendation quality. Web crawling and public datasets are used to collect food names, serving weights, energy, carbohydrates, protein, fat, sodium, ingredient composition, preparation methods, allergy information, and price ranges. Data expressed in different units are standardized to grams, and ingredients with similar names are mapped to a common standard name through normalization. The database separates foods, nutrients, ingredients, and tags so that new foods and ingredients can be added easily.

The meal-recommendation and repetition-checking module receives the user's daily nutrient targets and the food database as inputs and constructs breakfast, lunch, and dinner. The daily target is divided among meals, and candidate foods are selected by considering allergies, budget, and meal timing. A final plan is chosen by jointly evaluating how closely the nutrient total of each candidate combination matches the target, whether the same ingredients were used during the preceding several days, and whether the price is excessive. The system records the most recent date on which each ingredient was used and the minimum allowable

repeat interval. In this way, common diet foods such as chicken breast and sweet potatoes can be prevented from recurring on consecutive days.

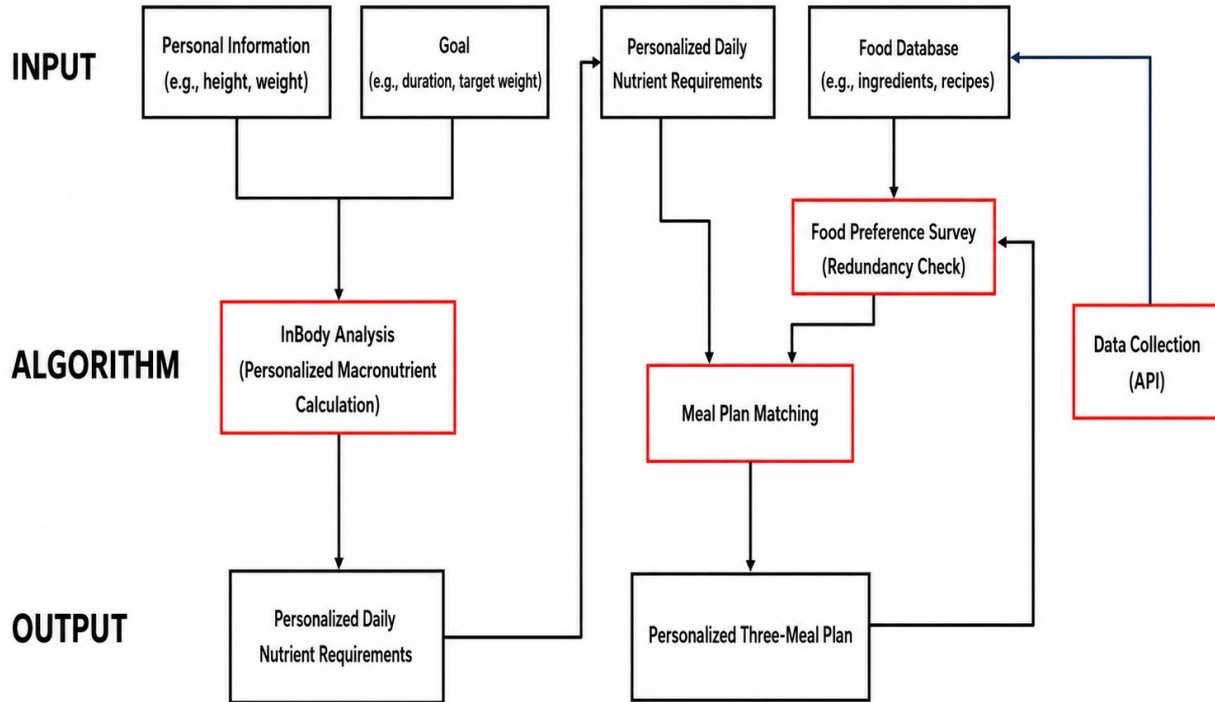


Figure 1. Flowchart of the Meal Recommendation System

Figure 1. Algorithm flow of the personalized meal recommendation system.

III. EXPERIMENT

The experiment examined whether the proposed meal-recommendation algorithm and repetition-checking logic could reliably construct three meals per day while maintaining a reasonable level of variety. A simulated database containing approximately 100 food items was created, and carbohydrate, protein, fat, and energy values were assigned to each item. Some foods were configured to share common ingredients so that ingredient repetition could be observed more readily.

One synthetic user was defined with daily targets of 375 g of carbohydrates, 80 g of protein, 70 g of fat, and 2,600 kcal. The proposed system was then used to automatically generate breakfast, lunch, and dinner for 30 days. From the generated plans, daily nutrient totals were calculated, and the intervals before the same menu item or principal ingredient reappeared were examined.

A 10-day sample of the 30-day results included a variety of breakfast items, such as a spinach bagel sandwich, sweet potato salad, soy milk oatmeal, a spinach frittata, and an oatmeal bowl. Lunch and dinner options were also distributed across different dishes, including a bulgogi rice bowl, chicken rice-paper rolls, shrimp and broccoli stir-fry, grilled mackerel, lentil salad, and pan-fried tofu. Even when the same menu item reappeared, it did so only after an interval of several days, indicating that the meal plans did not repeat monotonously.

Daily targets: Carbohydrates 375 g | Protein 80 g | Fat 70 g | Energy 2,600 kcal

Day	Breakfast	Lunch	Dinner	Carbs (g)	Protein (g)	Fat (g)	Calories (kcal)
1	Spinach Bagel Sandwich	Bulgogi Rice Bowl	Braised Potatoes	122	81.2	46	1226.8
2	Sweet Potato Salad	Chicken Rice-Paper Rolls	Pan-Fried Tofu	124.5	77.6	40.7	1174.7
3	Soy Milk Oatmeal	Chicken-Vegetable Stir-Fry	Bulgogi	92.5	74.3	45.8	1079.4
4	Spinach Frittata	Shrimp-Broccoli Stir-Fry	Grilled Mackerel	105	73.6	50	1164.4
5	Soy Milk Smoothie	Chicken-Tofu Pasta	Tomato Chicken Stir-Fry	124.2	79.9	34.9	1130.5
6	Brown Rice	Lentil Salad	Braised Lotus Root	105.6	77.3	46	1145.6
7	Broccoli Soup	Light Spicy Noodles	Pan-Fried Tofu	103.4	78.9	40.4	1092.8
8	Oatmeal Bowl	Beef-Vegetable Stir-Fry	Seasoned Bok Choy	126.1	73.8	34.9	1113.7
9	Spinach Bagel Sandwich	Eggplant Rice Bowl	Braised Potatoes	113.4	75.1	41.8	1130.2
10	Sweet Potato Salad	Bulgogi Rice Bowl	Soybean Paste Stew	92.2	87.1	39.7	1074.5

Figure 2. Ten-day example generated by the meal-matching algorithm.

The daily totals for carbohydrates, protein, fat, and energy were somewhat below their respective targets, but day-to-day variation was not excessive and remained within a relatively consistent range. At this stage, the result indicates that the algorithm can identify reasonable combinations while considering both nutritional balance and meal variety. Compared with operation without the repetition-checking function, the proportion of ingredients reused on consecutive days decreased. This reduction could help mitigate menu fatigue in actual use.

IV. CONCLUSION

This study proposed a system that automatically recommends personalized meal plans using InBody-based body-composition data. The system architecture and core algorithms were designed, and their feasibility was examined through simulation. By moving beyond simple weight-centered management and using detailed indicators such as skeletal muscle mass, body-fat percentage, and BMR, the proposed system can provide more precise nutritional targets.

The system estimates BMR and total daily energy expenditure from either InBody measurements or basic physical information and then incorporates the user's objective and activity level to establish recommended daily energy, carbohydrate, protein, and fat intake. This process allows users to receive meal plans that are more closely aligned with their physical condition. In addition, web-based data collection and a normalized food-database design provide a foundation for systematically managing nutritional information for diverse foods.

The experiment confirmed that combining the meal-recommendation algorithm with ingredient-repetition checking allowed the system to maintain a consistent nutritional balance while preventing principal ingredients and menu combinations from recurring excessively over short periods. This feature may reduce boredom and improve the practical sustainability of long-term dietary and body-composition management.

Future work may integrate an actual InBody API to collect body-composition data automatically and expand the food database to include branded products and restaurant menus. If users' actual intake records and longitudinal body-composition changes are collected and incorporated into a machine-learning-based recommendation model, more sophisticated personalized meal recommendations and health-status predictions may become possible.

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